Object-Grounded Text Priors for Saliency Ranking Detection

Anonymous WACV Algorithms Track submission

Paper ID 365

Abstract

Saliency ranking detection (SRD) identifies multiple salient objects and orders them by relative visual importance. Existing methods mainly infer this hierarchy from visual appearance and inter-object reasoning, while recent language-guided approaches use descriptions largely as global scene context or caption-derived relations. We argue that language can more effectively complement visual evidence for SRD when object-referential sentences guide saliency ranking at the object level. We propose ***TSPNet****, a framework that converts detailed image descriptions into object-level auxiliary priors for saliency ranking. Without direct sentence–object saliency annotations, TSPNet learns text-conditioned auxiliary scores for CLIP-matched ROI–sentence pairs using human-annotated saliency ranks as indirect ordinal supervision. It renders these scores as spatial heatmap priors, while the corresponding text representations guide ROI feature refinement. This extends language guidance beyond global caption modulation toward ROI-specific saliency reasoning. Experiments on ASSR, IRSR, and SIFR show consistent improvements over strong recent baselines in the primary ranking metrics, SA-SOR and SOR. Ablations further show that these gains depend on meaningful linguistic cues and correct ROI–sentence correspondence.*

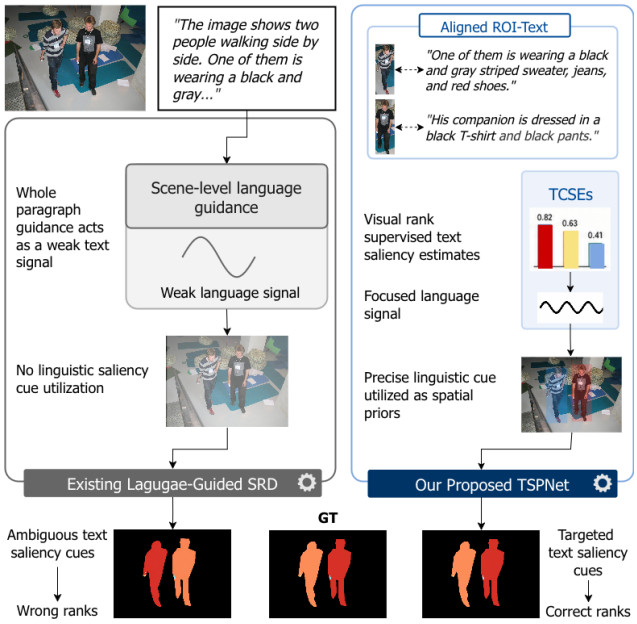


Figure 1. Comparison of language usage in existing language-guided SRD methods and TSPNet. Existing methods exploit descriptions as global scene-level guidance or caption-derived semantic cues, which can weaken object-specific saliency signals. In contrast, TSPNet grounds object-referential sentences to individual ROIs and converts them into text-conditioned saliency priors, enabling focused object-level guidance for saliency ranking.

1. Introduction

Saliency ranking detection (SRD) aims to segment multiple objects in an image and order them according to their relative visual importance. By identifying not only which objects are salient but also how they compete for attention, SRD provides structured object-level cues for downstream vision tasks. In visual question answering and image captioning, ranking salient objects can help models focus on informative regions and generate more accurate responses [5, 6]. It also supports applications that require selective allocation of visual attention, such as visual retrieval [21, 31] and image compression [22].

Recent progress in SRD has been largely driven by advances in visual representation learning. Early methods inferred object importance from object-level [9, 27] or pixel-level [12] visual features, often treating objects independently. Later approaches [8, 16, 30] introduced contextual reasoning by modeling interactions among salient objects. Despite these advances, most SRD methods remain predominantly vision-centric, relying mainly on visual appearance, spatial layout, and object-level context.

While saliency ranking is inherently grounded in visual perception, natural-language image descriptions may provide complementary saliency-correlated cues, such as which objects are mentioned, how they are emphasized,

and how much descriptive detail they receive. This motivates a key question: can such language be converted into useful auxiliary priors for visual saliency ranking? Existing language-guided SRD methods, such as LG-SOR [15], show that large vision-language model (LVLM) generated descriptions can improve saliency ranking through text-guided backbone modulation and entity-relation reasoning. However, these methods primarily use language as global semantic context for backbone modulation or as caption-derived relational cues between objects, leaving the use of object-referential language for SRD underexplored. Our method addresses this gap by learning text-conditioned saliency scores under ordinal supervision from ground-truth saliency ranks. It injects these scores as spatial priors and uses the corresponding text representations for feature-level refinement. Figure 1 summarizes the distinction between our method and existing language-guided SRD models.

In this study, we propose **T**ext-Derived **S**patial **P**rior **Net**work (**TSPNet**), a saliency-ranking framework that learns text-derived auxiliary priors from object-referential sentences CLIP-matched to individual object regions of interest (ROIs) [25]. For each matched ROI–sentence pair, our text-conditioned saliency estimate (TCSE) module integrates the ROI representation with token-level sentence representations to predict an auxiliary TCSE score. Because direct sentence–ROI saliency annotations are unavailable, these scores are trained with pairwise ordinal supervision from ground-truth saliency ranks: a higher-ranked ROI is encouraged to receive a score at least a predefined margin below a lower-ranked ROI. We treat the resulting scores as noisy, complementary cues rather than direct measures of visual saliency. They are rendered into image space as heatmap overlays, providing the visual backbone with explicit text-derived spatial priors. In addition, the learned text representations guide multi-kernel refinement of the corresponding ROI features according to their matched descriptions.

Because existing SRD datasets do not contain both saliency ranking annotations and long-form image descriptions, we leverage large vision-language models (LVLMs) to generate detailed image descriptions. Compared with conventional short captions from datasets such as MS COCO [14] and Flickr30K [23], these descriptions provide richer linguistic cues, enabling more informative use of language for saliency ranking. Extensive experiments across multiple SRD benchmarks demonstrate that incorporating object-referential textual cues improves saliency ranking accuracy and produces predictions that better align with human annotations.

- We introduce a text-guided SRD framework that learns auxiliary spatial priors from CLIP-matched ROI–sentence pairs using ordinal supervision from ground-truth saliency ranks, moving beyond global caption modulation toward explicit ROI-specific textual guidance.
- We propose TCSE, an indirectly supervised module that predicts auxiliary text-conditioned scores for CLIP-matched ROI-sentence pairs and converts them into spatial priors. Without direct ROI-sentence saliency annotations, TCSE is trained with pairwise supervision so that the score ordering follows the annotated visual saliency ranks.
- We design a text-adaptive multi-kernel feature extraction pathway that uses learned text representations to condition convolutional filtering over ROI features, and show through extensive experiments and ablations that object-referential linguistic cues improve SRD performance across multiple benchmarks.

2. Related Work

2.1. Saliency Ranking

Most prior work on saliency ranking detection (SRD) formulates the task as a vision-centric inference problem, estimating relative object importance from appearance cues, spatial configuration, and inter-object interactions. Early work such as RSDNet [12] jointly addresses salient object detection and ranking by learning saliency representations that implicitly encode relative prominence. ASSR [27] further models attention shifts between objects to infer ranking order. SOR [9] proposes position-preserved attention to model object importance, while OCOR [30] emphasizes bi-directional object-context prioritization to refine ranking under cluttered scenes. IRSR [16] introduces graph reasoning over object representations to capture contextual dependencies among instances. HyperSOR [24] uses a graph hypernetwork to incorporate scene context into ranking. DSGNN [32] models ranking relationships through shape and texture graphs. QAGNet [8] advances SRD by taking a hierarchical approach to saliency modeling. Despite these advances, existing approaches rely predominantly on visual evidence and largely overlook how linguistic cues can complement visual features for object-level saliency ranking.

2.2. Auxiliary Cues and Language Guidance

To better approximate human attentional behavior, some approaches incorporate auxiliary priors beyond generic appearance features. PoseSOR [10] leverages human pose to enhance ranking, motivated by the observation that attention in human-centric scenes is shaped by body configuration and human-object interactions. More recently, language has begun to be explored as an additional source of context for SRD [26, 29]. However, these methods typically focus on pixel-level saliency prediction.

LG-SOR [15] uses LVLM-generated descriptions to enrich visual reasoning, extracting caption-level semantic en-

tities and relations to support object interaction modeling. This shows that descriptive language can provide useful information beyond visual appearance alone. However, its linguistic signal primarily serves as global semantic context for backbone modulation and relation-level supervision between objects; it does not incorporate textual information specific to individual ROIs. More broadly, existing methods have not fully explored how object-referential sentences can guide saliency ranking through spatial prior injection and text-adaptive ROI feature refinement. In contrast, our method learns an auxiliary score for each CLIP-matched ROI–sentence pair under ordinal supervision from ground-truth saliency ranks. These scores are converted into spatial heatmap priors, while the corresponding text representations guide multi-kernel refinement of ROI features.

3. Proposed Method

Pipeline Overview. Our goal is to use object-referential language to guide saliency ranking at both the input and feature levels. Specifically, we derive ordinally supervised spatial priors for input-level guidance and use the corresponding text representations to refine ROI features. Given an RGB image $I \in \mathbb{R}^{H\times W\times 3}$ and an LVLM-generated description L, a Mask2Former [4]-based segmentation branch predicts N instance masks that define the candidate ROIs $R = \{r_k\}_{k=1}^{N}$. The description is decomposed into sentences $S = \{s_j\}_{j=1}^{M}$, and each ROI is matched to its most similar sentence using cosine similarity in a frozen CLIP vision-language embedding space [25].

For each CLIP-matched ROI-sentence pair (r_k, s_j), the Text-Conditioned Saliency Estimate (TCSE) module encodes the sentence and cross-attends it with its corresponding ROI to estimate an ROI-level text-conditioned saliency prior, referred to as the TCSE score. Since direct sentence–object saliency annotations are unavailable, the TCSE module is trained with a pairwise ranking objective using ground-truth saliency ranks as indirect ordinal supervision. The resulting TCSE scores are rendered onto the image as a gated heatmap overlay $O \in \mathbb{R}^{H\times W\times 3}$, producing an overlay-augmented image I' that provides the backbone with explicit text-derived spatial guidance.

The same learned text representations guide the Multi-Kernel Text-Adaptive (MKTA) Feature Extractor in refining multi-scale ROI features. ROIAlign extracts backbone features for ROIs defined by the segmentation branch's predicted masks. For each r_k, MKTA uses their matched sentence s_j to generate text-conditioned convolutional kernels with multiple receptive fields and applies them to the ROI features. The resulting text-adaptive features are fused across paths and scales, then passed to a scene-context reasoning module to produce context-aware ROI features. The rank head uses these ROI features to predict object-level saliency scores. The segmentation branch is first trained separately using backbone-generated full-image feature maps and is then frozen during saliency training. It provides the predicted masks used to define the ROIs and to construct the final saliency map. The final saliency map $\hat{Y} \in \mathbb{R}^{H\times W}$ is obtained by composing the predicted masks weighted by their saliency scores.

3.1. Text-Conditioned Saliency Estimate (TCSE) Module

Given candidate object regions and description sentences, the TCSE module predicts an auxiliary text-conditioned saliency score for each CLIP-matched ROI-sentence pair. The module is trained with pairwise visual-rank supervision, and the resulting scores are rendered as spatial priors.

A frozen CLIP encoder maps the ROIs $\{r_k\}$ and image-description sentences $\{s_j\}$ into a shared embedding space, producing $\{r_k^{clip}\}$ and $\{s_j^{clip}\}$, respectively. For each ROI r_k, we select the sentence s_j with the highest cosine similarity: $j = \arg\max_{\ell\in\{1,\dots,M\}} \cos_\mathrm{sim}(r_k^{clip}, s_\ell^{clip})$. To prevent unrelated text from influencing prediction, we select $\tau = 0.2$ using the validation split and fix it for all test-time inference. If the maximum similarity for an ROI is below τ, we assign it an empty sentence $s_j =$ "". Because ROI crops lack global image position, CLIP matching relies primarily on crop-visible linguistic cues such as categories and attributes, while spatial terms such as "left" and "right" provide limited matching cues. s_j is tokenized using the CLIP tokenizer and processed by a learnable text encoder, which adds positional encodings to the token embeddings and applies self-attention, residual connections with normalization, and a feed-forward network (FFN), yielding $s_j^{enc} \in \mathbb{R}^{T_j\times d}$. For each matched pair (r_k, s_j), cross-attention uses r_k^{clip} as the query and s_j^{enc} as the keys and values. Its attended output is projected to a normalized TCSE score $e_{k,j} \in [0, 1]$. Each score is mapped to an RGB color C_k through piecewise-linear interpolation over fixed heatmap anchors and rendered over r_k to form overlay O, which is added to the image through a learned gate γ_o to produce I'.

The TCSE module is trained with a *Pairwise Ranking Loss* that uses ground-truth visual ranks as indirect ordinal supervision. Let $e^i_{k,ms(k)}$ denote the TCSE score for ROI k with its matched sentence $ms(k)$ in image i, and let $\rho_i(k)$ denote its ground-truth visual saliency rank, where a smaller rank indicates higher saliency. For each image, we construct the ordered pair set $\mathcal{P}_i = \{(u, v) \mid \rho_i(u) < \rho_i(v)\}$, where ROI u is ranked above ROI v according to the ground-truth saliency ranks. The TCSE ranking loss is defined as

$$\mathcal{L}_{\mathrm{TCSE}} = \frac{1}{\sum_i |\mathcal{P}_i|} \sum_i \sum_{(u,v)\in\mathcal{P}_i} \Big[m - \big(e^i_{v,ms(v)} - e^i_{u,ms(u)}\big)\Big]_+, \tag{1}$$

where $[x]_+ = \max(0, x)$ and m is the ranking margin.

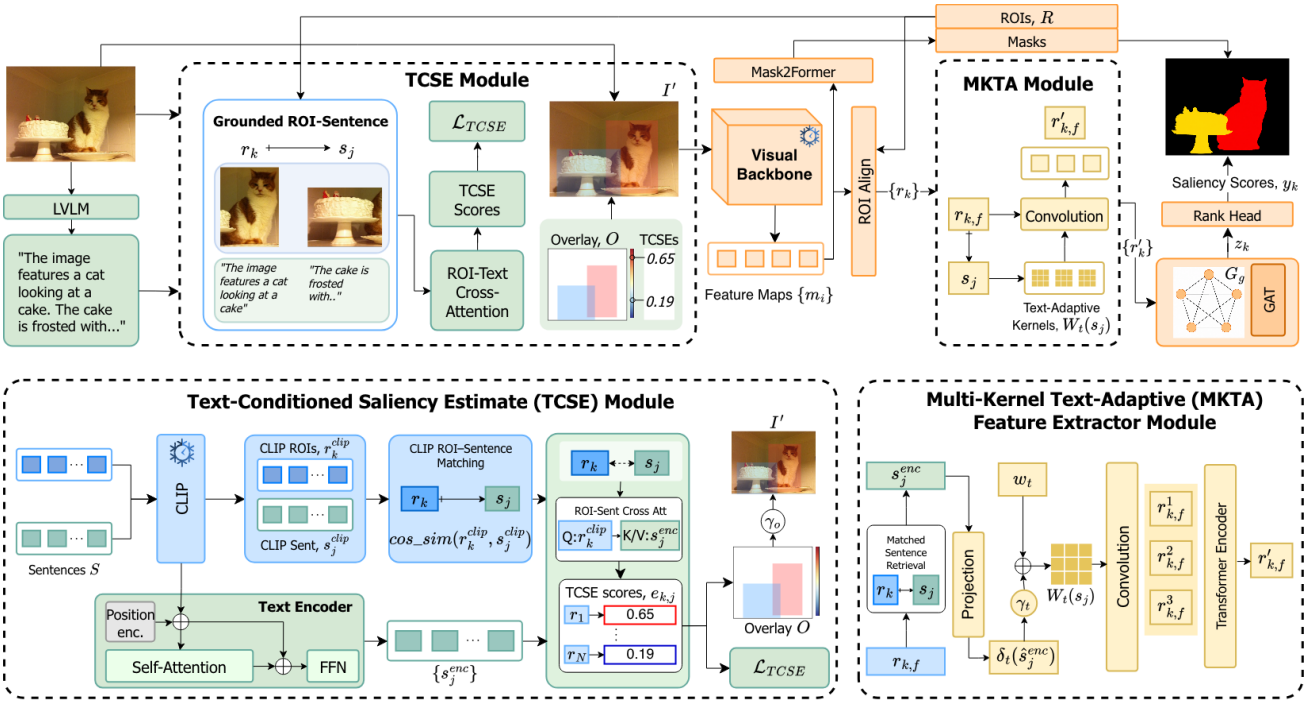


Figure 2. Overview of TSPNet. Given an image and its object-referential description, TSPNet incorporates language into saliency ranking through three steps: (i) CLIP-matching object-referential sentences to candidate ROIs, (ii) learning text-conditioned auxiliary scores using ordinal supervision from ground-truth saliency ranks and rendering them as spatial heatmap priors, and (iii) using the corresponding text representations for multi-kernel ROI feature refinement. The refined features are combined with scene-context reasoning to predict object-level saliency ranks, providing ROI-specific textual guidance beyond global caption modulation.

This loss encourages the TCSE score of a visually higher-ranked ROI to be lower than that of a lower-ranked ROI by at least m, so that TCSE scores preserve the relative visual saliency order within each image. This objective does not assume that TCSE scores are direct measures of visual saliency. Instead, image descriptions are treated as noisy but complementary saliency-correlated cues. Visual saliency ranks provide indirect ordinal supervision, training the TCSE module to order the TCSE scores of matched ROI-sentence pairs according to relative visual importance. In the TCSE module, ROIs without a sufficiently similar sentence are paired with an empty sentence, which is processed like other matches so that every ROI receives a TCSE score.

3.2. Multi-Kernel Text-Adaptive Feature Extractor

The Multi-Kernel Text-Adaptive (MKTA) Feature Extractor refines ROI features using their matched sentences. The TCSE module uses a fixed colormap to encode TCSE scores as spatially localized, three-channel ROI priors, allowing standard RGB backbones to jointly process text-derived saliency cues and visual appearance without architectural modifications. Given the overlay-augmented image I', the backbone produces multi-scale feature maps $\{m_f\}$, from which ROIAlign uses the segmentation branch's predicted masks to extract $r_{k,f}$ for each ROI r_k. The encoded matched sentence s_j^{enc} is projected to $\hat{s}_j^{enc}$. For each convolutional path t, δ_t converts this representation into an offset for the learned base kernel $W_t(s_j^{enc}) = w_t + \gamma_t \delta_t(\hat{s}_j^{enc})$, where the learned gate γ_t controls the text contribution. If an ROI has no matched sentence after τ-filtering, the offset is set to zero and the path uses only w_t. This residual design lets the stable base kernel w_t extract features from ROIs lacking matched text.

MKTA applies three parallel text-adaptive convolutions to the same ROI feature $r_{k,f}$, using kernels $c_1 = (3\times3, \text{dil} = 1)$, $c_2 = (3\times3, \text{dil} = 2)$, and $c_3 = (5\times5, \text{dil} = 2)$. Kernels of different receptive fields allow MKTA to capture local and broader context. A Transformer encoder fuses the three path responses $\{r_{k,f}^t\}$ into a scale-specific ROI representation $r'_{k,f}$. Cross-attention then combines $\{r'_{k,f}\}_f$ across backbone scales to obtain r'_k. SCB constructs a fully connected graph G_g with r'_k as node features and applies a single-layer Graph Attention Network (GAT), producing context-aware features z_k. The rank head predicts saliency scores y_k, which weight the masks produced by the separately trained segmentation branch to form the final saliency map $\hat{Y}$.

4. Experiments

4.1. Dataset and Metrics

We evaluate our method on three salient object ranking (SOR) benchmarks: IRSR [16], ASSR [27], and SIFR [8]. ASSR provides 7,464 training, 1,436 validation, and 2,418 test images, with five ranked salient objects per image. IRSR includes 6,059 training and 2,929 test images, where each image contains 1–8 ranked salient objects. SIFR consists of 6,701 training and 1,688 test images. All three datasets provide rank annotations for every annotated object in each image. Our method is evaluated against representative state-of-the-art baselines using three standard metrics that are commonly adopted in recent SOR literature: SOR [28], SA-SOR [16], and mean absolute error (MAE).

The SOR metric computes the average Spearman's rank correlation between the predicted saliency ordering of objects and the corresponding human-annotated rankings. Following prior work, this metric is evaluated only on images containing at least two overlapping objects, and the resulting scores are normalized to the range [0,1]. SA-SOR extends this evaluation by first matching predicted and ground-truth object instances using Intersection over Union (IoU) and then computing Spearman's rank correlation over the matched instances. Consistent with [8, 16], we adopt an IoU threshold of 0.5. Finally, MAE is computed as the mean pixel-wise absolute difference between the predicted saliency ranking maps and their corresponding ground-truth maps, providing a complementary measure of ranking accuracy at the pixel level.

Table 1. Description statistics.

Statistic	Value
Avg. sentences per description	6.2713
Avg. tokens per sentence	20.8225
Avg. tokens per description	130.5845
Max sentence length before truncation (tokens)	112
Percentage of sentences truncated	2.42%

4.2. Baseline Methods

We compare our method with seven recent saliency ranking approaches: RSDNet [12], ASSR [27], IRSR [16], SOR [9], OCOR [30], QAGNet [8], and LG-SOR [15]. Whenever available, we use the official pre-trained models released by the authors; otherwise, models are retrained using the publicly available source code and training protocols described in the original papers. Several methods predict a fixed number of salient objects. For RSDNet, ASSR, and SOR, we set this number to match the maximum number of objects used in their original implementations to ensure a fair comparison. LG-SOR results are reproduced using its best-performing caption generator according to [15], InstructBLIP [7].

4.3. Implementation Details

We use pretrained and frozen ResNet-50, ResNet-101 [11], Swin-B, and Swin-L [17] as image encoders, following common practice in prior SOR methods [8, 9, 12, 16, 27, 30]. All backbone weights are initialized from publicly available ImageNet-pretrained models. For ROI-text matching, we use the pretrained and frozen CLIP ViT-B/32 encoder released by OpenAI [25] to map visual regions and textual descriptions into a shared embedding space. We set the CLIP matching threshold to $\tau = 0.2$, selected using the validation split. All input images are resized to 1024×1024, consistent with the preprocessing protocol of [8]. The learnable text encoder has a depth of 1, and the hidden layer size is set to 512. TCSE scores are converted into RGB heatmap priors using piecewise-linear interpolation over four fixed color anchors: deep blue $(0.10, 0.05, 0.60)$, cyan $(0.00, 0.70, 0.95)$, yellow $(0.95, 0.90, 0.00)$, and red $(0.85, 0.05, 0.05)$, corresponding to normalized saliency levels 0.00, 0.33, 0.67, and 1.00, respectively.

During training, we first warm up the Mask2Former segmentation branch for 10,000 iterations using only training-split mask annotations, while keeping the saliency branch frozen. The trained segmentation branch then generates predicted ROIs for both saliency-branch training and evaluation. During saliency training, each predicted ROI is assigned the rank of its highest-IoU ground-truth instance if the IoU exceeds 0.5; otherwise, it is excluded from rank-supervision losses. Duplicate matches to the same ground-truth instance are resolved by keeping the highest-IoU prediction. The saliency branch is trained with the pairwise ranking loss [2] and the proposed TCSE ranking loss, while mask prediction uses Dice loss [20] and saliency classification uses binary cross-entropy [1]. The TCSE hinge loss margin m is set to 0.1 based on the validation set. All losses are assigned equal weights in the loss composition. Following [8], the final instance confidence is the product of saliency class confidence and mask confidence, and instances with confidence above 0.5 are retained. All trainable weights are optimized with AdamW [19], an initial learning rate of 1×10^{-5} decayed to 1×10^{-7} by cosine annealing [18], on two RTX 4090 GPUs with batch size 4.

4.3.1. Image Description Generation

All experiments use long-form descriptions generated by ShareGPT4V [3], unless specified otherwise. For each image, we use a fixed prompt template: *"Describe the image in detail within 200 words."* Descriptions are generated offline once and cached for both training and evaluation. Caption generation time using two NVIDIA GeForce RTX 4090 GPUs, with a batch size of 64, was recorded to be 0.30 seconds per caption. We generate descriptions

Table 2. Quantitative comparison with saliency-ranking methods using CNN backbones (ResNet [11] and VoVNet [13]) and Transformer backbones. TSPNet is evaluated with ResNet-50/101 [11] and Swin-B/L [17]. -U and -L denote unlimited and limited variants, respectively. Best and second-best results are shown in red and blue. **#Tr.** and **#Tt.** report trainable and total parameter counts, respectively. For TSPNet, **#Tr.** excludes frozen components, including the backbones, whereas **#Tt.** includes all saliency-inference components. For TSPNet and LG-SOR, **#Tt.** excludes the frozen LVLM because captions are generated and cached offline.

Method	Backbone	ASSR			IRSR			SIFR			Params(M)	
		SA-SOR↑	SOR↑	MAE↓	SA-SOR↑	SOR↑	MAE↓	SA-SOR↑	SOR↑	MAE↓	#Tr.	#Tt.
CNN-Based												
RSDNet [12]	ResNet-101	0.6313	0.7758	0.1236	0.4232	0.7096	0.1175	0.4791	0.7239	0.0772	42.7	42.7
ASSR [27]	ResNet-101	0.5400	0.7920	0.1010	0.3207	0.6521	0.1098	0.3281	0.5843	0.0624	44.2	44.2
IRSR-U [16]	ResNet-50	0.7051	0.8314	0.0923	0.5647	0.8143	0.0953	0.5585	0.7487	0.0465	128.1	128.1
IRSR-L [16]	ResNet-50	0.7090	0.8283	0.0914	0.5648	0.8141	0.0953	0.5585	0.7487	0.0465	128.1	128.1
SOR [9]	VoVNet-39	0.6371	0.8330	0.0799	0.5171	0.7909	0.0988	0.3820	0.7554	0.0580	119.0	119.0
QAGNet [8]	ResNet-50	0.7545	0.8514	0.0619	0.6110	0.8108	0.0845	0.6119	0.7899	0.0437	47.3	47.3
LG-SOR [15]	ResNet-50	0.7360	0.8875	0.0650	0.5690	0.8150	0.0620	0.5840	0.7900	0.0380	46.8	46.8
Ours (Res-50)	ResNet-50	0.7975	0.9002	0.0913	0.7009	0.8521	0.0521	0.6151	0.8059	0.0470	73.4	97.2
Ours (Res-101)	ResNet-101	0.8117	0.9071	0.0621	0.7144	0.8613	0.0490	0.6208	0.8143	0.0304	73.4	118.2
Transformer-Based												
OCOR [30]	Swin-L	0.6413	0.8843	0.0786	0.5183	0.8149	0.1003	0.4392	0.7436	0.0488	401.7	401.7
QAGNet-SwinB [8]	Swin-B	0.7741	0.8583	0.0538	0.6252	0.8152	0.0792	0.6167	0.7933	0.0409	110.2	110.2
QAGNet-SwinL [8]	Swin-L	0.7793	0.8591	0.0492	0.6466	0.8241	0.0768	0.6206	0.7982	0.0416	218.8	218.8
LG-SOR [15]	Swin-L	0.7810	0.8920	0.0490	0.6390	0.8410	0.0480	0.6130	0.8040	0.0370	218.3	218.3
Ours (Swin-B)	Swin-B	0.8146	0.9057	0.0480	0.7066	0.8551	0.0406	0.6177	0.8125	0.0279	73.4	161.7
Ours (Swin-L)	Swin-L	0.8142	0.9104	0.0459	0.7214	0.8615	0.0355	0.6240	0.8140	0.0277	73.4	292.0

with temperature 0.2, top-p sampling disabled, beam size 1, and a maximum length of 512 new tokens. The LVLMs are frozen in all experiments. Table 1 reports the description statistics after sentence segmentation. Since the CLIP text encoder [25] supports a maximum input length of 77 tokens, sentences exceeding this limit are truncated. Only 2.42% of sentences require truncation, indicating that this preprocessing step has minimal overall impact.

4.4. Performance Comparison

4.4.1. Quantitative Comparison

Table 2 shows that our method achieves state-of-the-art SA-SOR and SOR performance in most evaluated settings, with competitive MAE across datasets and backbone architectures. These improvements are observed across CNN-based and Transformer-based settings, including ResNet-50/101 and Swin-B/L variants of our model, indicating strong generalization across different visual feature extractors.

All percentages reported below are relative improvements, computed as $100 \times (\text{ours} - \text{baseline})/\text{baseline}$. Among CNN-based methods, our ResNet-101 variant improves SA-SOR over the strongest baseline, QAGNet, by 7.58% on ASSR, 16.92% on IRSR, and 1.45% on SIFR. For SOR, it outperforms the strongest baseline, LG-SOR, by 2.21% on ASSR, 5.68% on IRSR, and 3.08% on SIFR. A similar trend is observed among Transformer-based methods. Our best-performing variant for each dataset improves SA-SOR by 4.30% over LG-SOR on ASSR and by 11.57% and 0.55% over QAGNet on IRSR and SIFR, respectively. Our Swin-L variant improves SOR over LG-SOR by 2.06% on ASSR, 2.44% on IRSR, and 1.24% on SIFR. Since both the CLIP encoder for ROI-text matching and visual backbones are frozen, our model has a consistent number of trainable parameters across all backbones. Overall, our method achieves the best SA-SOR/SOR on most evaluated settings and competitive MAE, with particularly large gains on IRSR. This shows that CLIP-matched ROI text provides useful cues for saliency ranking. These cues complement visual evidence through spatial prior injection and text-adaptive feature refinement.

4.4.2. Qualitative Comparison

Figure 3 compares TSPNet with representative baselines. TSPNet produces rankings that more closely follow the ground-truth saliency order, especially in scenes with visually similar or spatially competing objects. In image (a), it

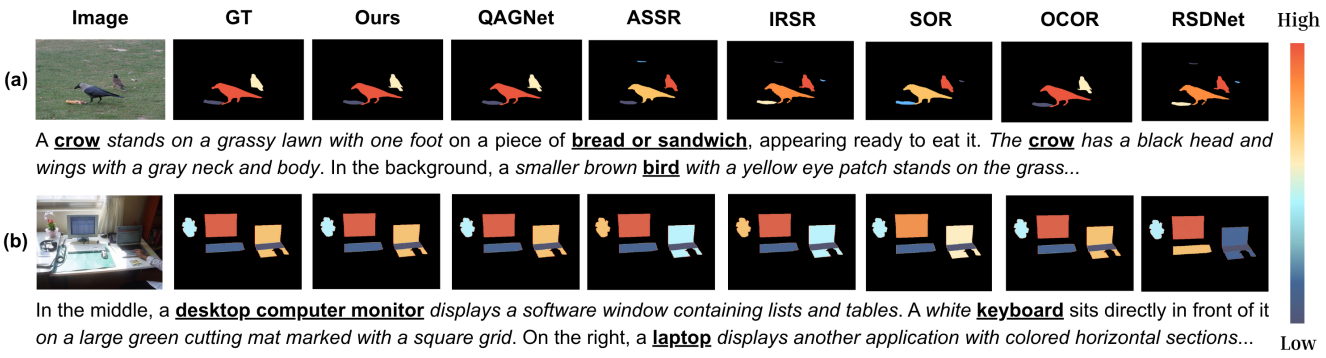


Figure 3. Qualitative comparison between our proposed TSPNet and other SRD methods on samples from the IRSR dataset

correctly prioritizes the visually dominant foreground crow over the smaller background bird by combining visual evidence with its object-referential description. In image (b), matched textual cues help the model correctly rank the monitor first and the laptop second, whereas other models rank the vase above the laptop. These examples suggest how object-referential language can provide additional object-level context in cases where visual cues alone may be ambiguous.

5. In-Depth Analysis

We analyze TSPNet through component ablations and perform diagnostic ablations on text perturbations, overlay perturbations, loss settings, and CLIP-matched ROI-text pairs. Unless otherwise specified, all experiments use IRSR dataset with ResNet-101 and ShareGPT4V captions.

5.1. Component Analysis

Table 3 evaluates the contribution of each component in TSPNet. Variant I is the visual baseline, where backbone features are directly passed to the ranking head. Variant II adds a full caption conditioned uniform overlay, where the full caption is encoded by the text encoder and projected to a single global TCSE score rendered over all ROIs; its small gain suggests that global linguistic context is limited without object-specific association. Variant III introduces ROI-conditioned sentence attention (RCSA). For each ROI, cross-attention learns weights over all sentences and combines their representations to predict the TCSE score, rather than selecting a single sentence using CLIP. Variant IV instead uses CLIP to select the most similar sentence for each ROI and predict its TCSE score from the matched pair. The resulting improvement suggests that CLIP-based matching improves saliency rank prediction. Variant V shows that TCSE scores become more effective with $\mathcal{L}_{\text{TCSE}}$ supervision. The full MKTA module in Variant VIII further improves results, demonstrating the benefit of text-adaptive multi-kernel ROI feature refinement. Finally, SCB yields the best performance, showing the benefit of contextual ROI modeling.

Table 3. Component analysis of the proposed model. **FD**: Full caption conditioned uniform overlay; **RCSA**: ROI-conditioned sentence attention over all sentences (no CLIP matching); **CLIP**: CLIP-based ROI-sentence matching; **TCL**: TCSE-score supervision via TCSE ranking loss $\mathcal{L}_{\text{TCSE}}$; **SK**: single-kernel feature extraction; **MK**: multi-kernel feature extraction; **TA**: text-adaptive feature refinement; **SCB**: Scene Context Blender.

Variant	TCSE				MKTA			SCB	SA-SOR ↑	SOR ↑
	FD	RCSA	CLIP	TCL	SK	MK	TA			
I									0.6555	0.8338
II	✓								0.6593	0.8368
III	✓	✓							0.6612	0.8385
IV	✓	✓	✓						0.6711	0.8411
V	✓	✓	✓	✓					0.6979	0.8519
VI	✓	✓	✓	✓	✓				0.6979	0.8519
VII	✓	✓	✓	✓	✓	✓			0.7045	0.8544
VIII	✓	✓	✓	✓	✓	✓	✓		0.7095	0.8588
IX	✓	✓	✓	✓	✓	✓	✓	✓	**0.7144**	**0.8613**

5.2. Diagnostic Ablations

Text perturbations. Table 4 evaluates the effect of perturbing the textual signal.The empty-caption setting retains TCSE processing with empty sentences while removing MKTA text-conditioned offsets, the image-shuffled caption setting replaces each image caption with a caption from another image, and sentence-mixed caption replaces a fraction of sentences with sentences sampled from other images. The 20% sentence-mixed setting tests partial corruption of sentence-level coherence, while the 100% setting breaks both image-text correspondence and within-caption consistency. Comparing these settings with full captions tests whether gains come from meaningful, image-specific textual content rather than merely from having text.

Overlay perturbations. Table 4 examines whether the TCSE overlay improves saliency ranking through correct TCSE priors, or simply by adding visual markers over object regions. The RGB overlay-only setting performs far

Table 4. Diagnostic ablations on textual cues, overlay injection, loss analysis, and CLIP-matched ROI-text. ρ denotes the Spearman correlation between TCSE scores and ground-truth ranks.

Setting	SA-SOR↑	SOR↑	
Text Perturbations			
Empty captions	0.6977	0.8424	
Sentence-mixed (20%)	0.7022	0.8542	
Sentence-mixed (100%)	0.6917	0.8488	
Image-Shuffled	0.6937	0.8490	
Full Caption	**0.7144**	**0.8613**	
Overlay Perturbations			
Image Only	0.6949	0.8502	
RGB Overlay Only	0.3788	0.6939	
Image + Scalar Prior	0.6845	0.8471	
Image + Shuffled RGB	0.6985	0.8519	
Image + Uniform RGB	0.7001	0.8526	
Image + Uniform Grayscale	0.6980	0.8519	
Image + TCSE Grayscale	0.7100	0.8588	
Image + TCSE RGB	**0.7144**	**0.8613**	
Loss Analysis			
$\mathcal{L}_{\text{TCSE}}$	0.3892	0.6872	
$\mathcal{L}_{\text{visrank}}$	0.6806	0.8425	
$\mathcal{L}_{\text{TCSE}} + \mathcal{L}_{\text{visrank}}$	**0.7144**	**0.8613**	
ROI-Text Matching Degradation			
0%	**0.7144**	**0.8613**	**0.44**
25%	0.7018	0.8538	0.40
50%	0.6845	0.8447	0.34
75%	0.6729	0.8407	0.30
100%	0.6671	0.8383	0.30

below all variants, showing that the overlay cannot replace visual evidence. Image + scalar prior also underperforms image-only, showing that non-spatial TCSE scores are insufficient. Shuffling TCSE scores modestly reduces performance, demonstrating the benefit of assigning each score to its corresponding ROI. However, results remain close to the unshuffled setting, indicating that TSPNet remains largely consistent with ground-truth ranks despite misleading TCSE priors. TCSE grayscale outperforms uniform grayscale, confirming that ROI-specific score variation contributes beyond spatial support. TCSE RGB performs best but only modestly exceeds TCSE grayscale, suggesting that score-aware spatial guidance provides most of the benefit, with RGB encoding offering a smaller additional gain.

Loss settings. Table 4 analyzes three loss settings while keeping the architecture and other training settings unchanged. With $\mathcal{L}_{\text{TCSE}}$ only, supervision is applied only to the TCSE scores, whereas with $\mathcal{L}_{\text{visrank}}$ only, supervision is applied only to the final predicted saliency ranks. The stronger performance of $\mathcal{L}_{\text{visrank}}$ confirms that it provides the primary ranking signal. Combining $\mathcal{L}_{\text{TCSE}}$ and $\mathcal{L}_{\text{visrank}}$ achieves the best performance, showing that auxiliary text-conditioned supervision complements the final visual-ranking objective.

Effect of ROI-Text Matching Degradation. Table 4 tests whether TSPNet relies on meaningful ROI-text correspondence by shuffling an increasing fraction of matched sentences across ROIs after CLIP-based matching. This preserves object-referential language but breaks its association with the correct object regions. As corruption increases, both the performance and TCSE-visual-rank correlation (ρ) drop. This shows that TSPNet benefits from correctly matched ROI-sentence pairs, and that TCSE scores depend on meaningful text-object correspondence rather than serving as direct visual-rank proxies. The fully shuffled setting performs worse than the image-shuffled setting because image-level shuffling often produces low-similarity CLIP matches that are filtered by the threshold τ and replaced with empty sentences, allowing the model to rely primarily on visual evidence. In contrast, ROI–text shuffling reassigns initially valid matches to incorrect ROIs, introducing misleading object-specific linguistic priors.

5.3. CLIP-Based ROI-Text Matching Reliability

Two authors independently evaluated CLIP-selected sentences for 301 IRSR ROIs. With $\tau = 0.2$, 300 ROIs received a retained match, yielding 99.67% coverage. Across 600 judgments of the retained matches, 78.17% were correct, with 85.00% inter-annotator agreement.

6. Conclusion

We presented TSPNet, a language-enhanced SRD framework that incorporates object-referential language through text-derived spatial priors and text-adaptive ROI feature refinement. For each CLIP-matched ROI–sentence pair, the TCSE module predicts an auxiliary score trained with a pairwise ranking loss that uses ground-truth saliency ranks as indirect ordinal supervision. These scores are rendered as spatial heatmap priors, while the corresponding text representations guide multi-kernel refinement of ROI features. This design enables descriptive language to support object-level saliency reasoning without direct sentence–object saliency annotations. Experiments on ASSR, IRSR, and SIFR show that TSPNet achieves state-of-the-art ranking performance and highly competitive MAE with both CNN and Transformer backbones. Ablations further indicate that the gains arise mainly from meaningful ROI-specific linguistic cues rather than generic text or simple colored ROI markers. Overall, our results suggest that CLIP-matched object-referential language provides useful cues for predicting saliency ranks. Future work will explore improved ROI–text matching, human-authored descriptions, and caption-noise robustness.

References

[1] Christopher M Bishop and Nasser M Nasrabadi. *Pattern recognition and machine learning*. Springer, 2006. 5

[2] Chris Burges, Tal Shaked, Erin Renshaw, Ari Lazier, Matt Deeds, Nicole Hamilton, and Greg Hullender. Learning to rank using gradient descent. In *ICML*, pages 89–96, 2005. 5

[3] Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Conghui He, Jiaqi Wang, Feng Zhao, and Dahua Lin. Sharegpt4v: Improving large multi-modal models with better captions. In *ECCV*, pages 370–387. Springer, 2024. 5

[4] Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *CVPR*, pages 1290–1299, 2022. 3

[5] Marcella Cornia, Lorenzo Baraldi, Giuseppe Serra, and Rita Cucchiara. Visual saliency for image captioning in new multimedia services. In *ICME*, pages 309–314. IEEE, 2017. 1

[6] Marcella Cornia, Lorenzo Baraldi, Giuseppe Serra, and Rita Cucchiara. Paying more attention to saliency: Image captioning with saliency and context attention. *IEEE TMM*, 14 (2):1–21, 2018. 1

[7] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale N Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning. *Advances in neural information processing systems*, 36:49250–49267, 2023. 5

[8] Bowen Deng, Siyang Song, Andrew P French, Denis Schluppeck, and Michael P Pound. Advancing saliency ranking with human fixations: Dataset models and benchmarks. In *CVPR*, pages 28348–28357, 2024. 1, 2, 5, 6

[9] Hao Fang, Daoxin Zhang, Yi Zhang, Minghao Chen, Jiawei Li, Yao Hu, Deng Cai, and Xiaofei He. Salient object ranking with position-preserved attention. In *CVPR*, pages 16331–16341, 2021. 1, 2, 5, 6

[10] Huankang Guan and Rynson WH Lau. Posesor: Human pose can guide our attention. In *ECCV*, pages 350–366. Springer, 2024. 2

[11] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016. 5, 6

[12] Md Amirul Islam, Mahmoud Kalash, and Neil DB Bruce. Revisiting salient object detection: Simultaneous detection, ranking, and subitizing of multiple salient objects. In *CVPR*, pages 7142–7150, 2018. 1, 2, 5, 6

[13] Youngwan Lee, Joong-won Hwang, Sangrok Lee, Yuseok Bae, and Jongyoul Park. An energy and gpu-computation efficient backbone network for real-time object detection. In *CVPR*, 2019. 6

[14] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *ECCV*, pages 740–755. Springer, 2014. 2

[15] Fang Liu, Yuhao Liu, Ke Xu, Shuquan Ye, Gerhard Petrus Hancke, and Rynson WH Lau. Language-guided salient object ranking. In *CVPR*, pages 29803–29813, 2025. 2, 5, 6

[16] Nian Liu, Long Li, Wangbo Zhao, Junwei Han, and Ling Shao. Instance-level relative saliency ranking with graph reasoning. *IEEE TPAMI*, 44(11):8321–8337, 2021. 1, 2, 5, 6

[17] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *ICCV*, pages 10012–10022, 2021. 5, 6

[18] Ilya Loshchilov and Frank Hutter. Sgdr: Stochastic gradient descent with warm restarts. *arXiv preprint arXiv:1608.03983*, 2016. 5

[19] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017. 5

[20] Fausto Milletari, Nassir Navab, and Seyed-Ahmad Ahmadi. V-net: Fully convolutional neural networks for volumetric medical image segmentation. In *3DV*, pages 565–571. Ieee, 2016. 5

[21] Alex Papushoy and Adrian G Bors. Image retrieval based on query by saliency content. 36:156–173, 2015. 1

[22] Yash Patel, Srikar Appalaraju, and R. Manmatha. Saliency driven perceptual image compression, 2020. 1

[23] Bryan A Plummer, Liwei Wang, Chris M Cervantes, Juan C Caicedo, Julia Hockenmaier, and Svetlana Lazebnik. Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to-sentence models. In *CVPR*, pages 2641–2649, 2015. 2

[24] Minglang Qiao, Mai Xu, Lai Jiang, Peng Lei, Shijie Wen, Yunjin Chen, and Leonid Sigal. Hypersor: Context-aware graph hypernetwork for salient object ranking. *IEEE TPAMI*, 46(9):5873–5889, 2024. 2

[25] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *ICML*, pages 8748–8763. PmLR, 2021. 2, 3, 5, 6

[26] Vasili Ramanishka, Abir Das, Jianming Zhang, and Kate Saenko. Top-down visual saliency guided by captions. In *CVPR*, pages 7206–7215, 2017. 2

[27] Avishek Siris, Jianbo Jiao, Gary KL Tam, Xianghua Xie, and Rynson WH Lau. Inferring attention shift ranks of objects for image saliency. In *CVPR*, pages 12133–12143, 2020. 1, 2, 5, 6

[28] Charles Spearman. The proof and measurement of association between two things. 1961. 5

[29] Yinan Sun, Xiongkuo Min, Huiyu Duan, and Guangtao Zhai. How is visual attention influenced by text guidance? database and model. *IEEE TIP*, 2024. 2

[30] Xin Tian, Ke Xu, Xin Yang, Lin Du, Baocai Yin, and Rynson WH Lau. Bi-directional object-context prioritization learning for saliency ranking. In *CVPR*, pages 5882–5891, 2022. 1, 2, 5, 6

[31] Haoxiang Wang, Zhihui Li, Yang Li, Brij B Gupta, and Chang Choi. Visual saliency guided complex image retrieval. 130:64–72, 2020. 1

[32] Zijian Wu, Jun Lu, Jing Han, Lianfa Bai, Yi Zhang, Zhuang Zhao, and Siyang Song. Domain separation graph neural networks for saliency object ranking. In *2024 IEEE/CVF*

Conference on Computer Vision and Pattern Recognition (CVPR), pages 3964–3974, 2024. 2